

PHASE 1 – BRAINSTORMING AND IDEATION

1. PROJECT TITLE

Import Data using Transform Maps (Spreadsheets)

2. PROJECT OVERVIEW

The proposed project focuses on implementing data import automation in the ServiceNow platform using Import Sets and Transform Maps. The system is designed around an Institution Details table (u_institution_details) containing student information such as student roll number, student name, faculty name, branch, email, phone number, and description. The primary data import requirement is to parse bulk student records from external spreadsheet files (Excel/CSV) into a staging table (u_institution_details_import) and automatically transform, validate, and populate them into target records within ServiceNow. In addition to bulk creation, field mapping rules, reference field resolution (sys_user), choice handling (ECE, EEE, CSE), and Coalesce duplicate prevention logic are established to ensure data integrity.

3. PROBLEM IDENTIFICATION

Organizations and educational institutions frequently collect bulk student and faculty data in external spreadsheets that must be transferred into ServiceNow. Manual entry of hundreds of student records is time-consuming, highly prone to human error, and inefficient for administrative workflows. Without an automated import set transformation mechanism, institutions face:

- Excessive manual effort and administrative overhead.
- Unvalidated reference records and broken user mapping.

- Inconsistent choice values for fields like Branch.
- Duplicate record creation upon subsequent updates.

Therefore, an automated mechanism using ServiceNow Transform Maps is required to upload spreadsheets, map fields, validate user references, handle choices, and prevent duplicates during bulk data loads.

4. PROPOSED SOLUTION

The proposed solution utilizes ServiceNow Import Sets and Transform Maps to execute automated bulk data imports. A custom staging table named `u_institution_details_import` receives uploaded raw spreadsheet data. A Transform Map (Institution Details Transform Map) is configured with specific field mapping and transformation rules to route data into the target custom table (`u_institution_details`).

Key components of the solution include:

- **Staging Table (`u_institution_details_import`):** Temporarily holds raw incoming spreadsheet rows.
- **Field Mapping:** Pairs source spreadsheet columns to destination fields in `u_institution_details`.
- **Reference Lookup:** Automatically resolves student and faculty names against the User (`sys_user`) table.
- **Choice Mapping:** Maps incoming branch strings to target Choice values (ECE, EEE, CSE).
- **Coalesce Configuration:** Sets unique identifier fields (e.g., email) as coalesce keys to update existing records and prevent duplicates.

5. PROJECT OBJECTIVES

The primary objectives of the project are:

- To configure an Import Set and staging table for loading Excel/CSV files into ServiceNow.
- To build a Transform Map linking staging fields to the target Institution Details table (u_institution_details).
- To validate reference field resolution against existing ServiceNow users (sys_user).
- To implement choice field validation for Branch options (ECE, EEE, CSE).
- To configure Coalesce rules for duplicate record detection and updates.
- To ensure transformed records respect existing post-import security policies (ACLs).
- To demonstrate the practical execution and verification of Transform Maps in ServiceNow.

6. PROPOSED SYSTEM

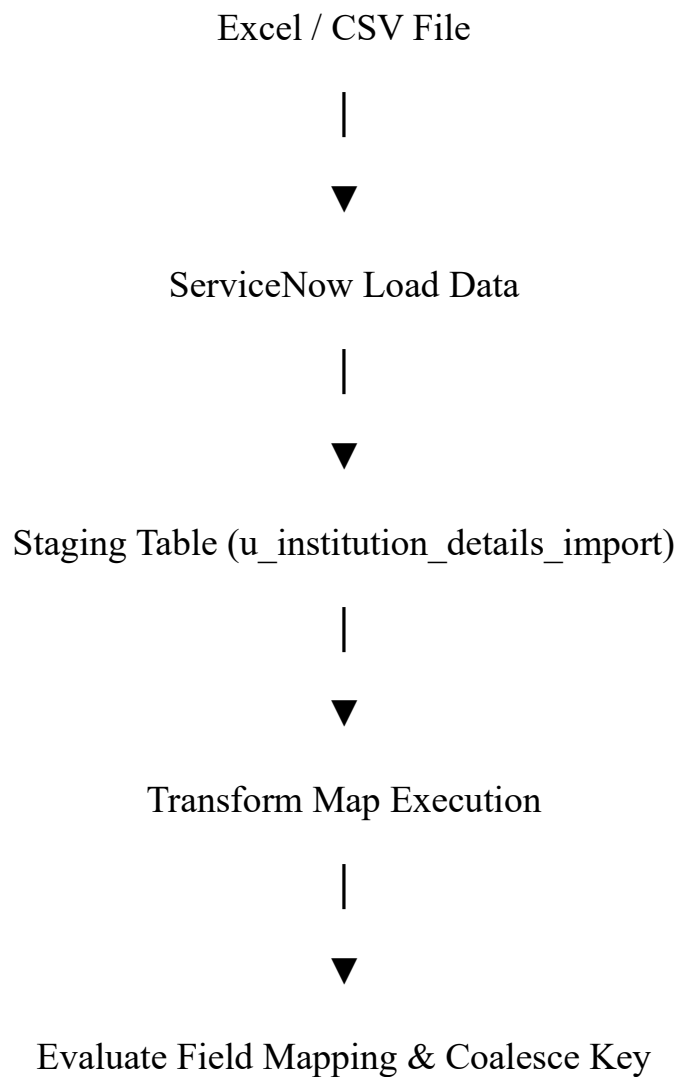
The proposed system consists of the following major components:

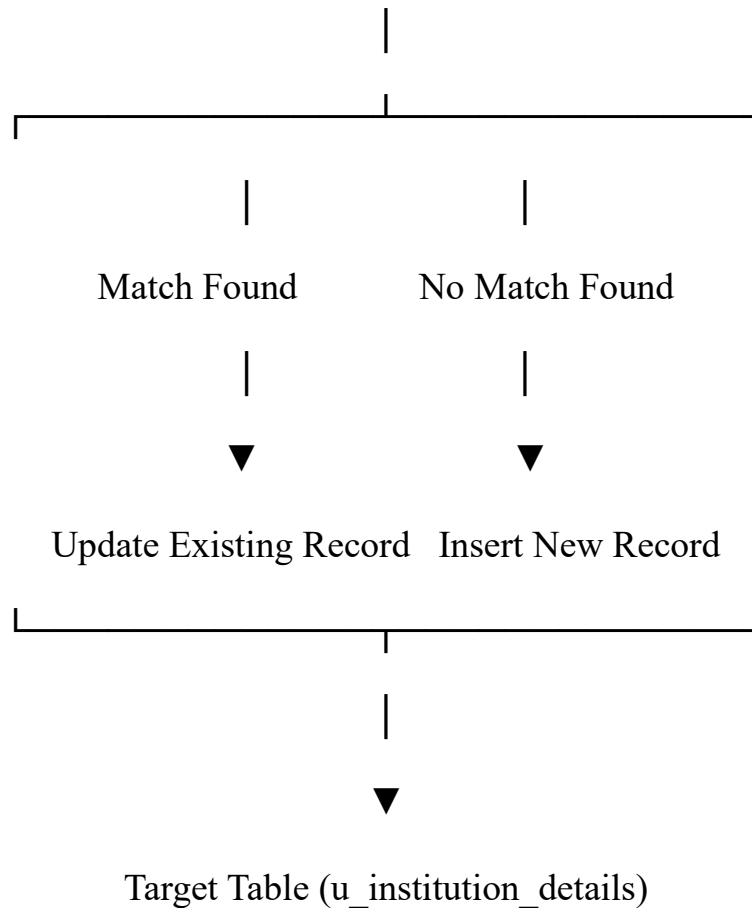
- **Data Preparation** A sample spreadsheet (Excel or CSV) is prepared containing fields for student roll numbers, names, faculty names, branches, emails, phone numbers, and descriptions.
- **Staging Table Creation** A custom staging table (u_institution_details_import) is generated via **System Import Sets > Load Data**.

- Transform Map Configuration A custom map (Institution Details Transform Map) defines source-to-target mapping relationships.
- Reference & Choice Mapping Rules Transformation scripts and field mapping rules validate string inputs into reference sys_ids (sys_user) and choice dropdown values (ECE, EEE, CSE).
- Import Set Verification The system is tested by transforming sample data rows and verifying row counts, error logs, and target record population in u_institution_details.

7. DATA IMPORT PROCESS MODEL

The overall data import workflow is represented below:





8.PROPOSED DATA STRUCTURE

The project uses the following custom target table structure:

Table Label: Institution Details **Table Name:** u_institution_details

Field	Type	Purpose / Mapping Rule
Student Roll Number	Auto Number	Unique identification (Auto-generated on insert)
Student Name	Reference – User	Identifies student (Mapped from sys_user ID/Name)

Field	Type	Purpose / Mapping Rule
Faculty Name	Reference – User	Identifies faculty member (Mapped from sys_user ID/Name)
Branch	Choice	Stores choice values (ECE, EEE, CSE)
Email	String	Stores contact email (Configured as Coalesce Key)
Phone Number	String	Stores phone contact details
Description	Multi-line String	Stores additional notes/information

9. TRANSFORM MAP SCENARIO

The project uses the following scenario to demonstrate Transform Map :

Bulk student records from different branches (ECE, EEE, CSE) are uploaded via spreadsheet. The system stages the file, maps columns to target attributes, validates references against sys_user, applies choice values, evaluates Coalesce matching on email, and populates u_institution_details.

Expected behavior:

Input Condition	Expected Result
New record entry	Inserts new record into u_institution_details
Matching email entry (Coalesce)	Updates existing record without creating duplicate
Valid user string	Resolves reference to corresponding sys_user record
Valid Branch string	Sets target Choice value to ECE, EEE, or CSE

10.PROJECT SCOPE

The scope of the project includes:

- Preparing source Excel / CSV data files.
- Loading source files into ServiceNow Import Sets.
- Creating the staging table (u_institution_details_import).
- Creating and verifying target table fields (u_institution_details).
- Configuring the Transform Map (Institution Details Transform Map).
- Defining field mappings, Mapping Assist rules, and Coalesce keys.
- Executing data transformations.

- Reviewing Import Set execution status and error logs.
- Verifying target table records and post-import security enforcement.
- Documenting the implementation steps and test results.

11.EXPECTED OUTCOME

The expected outcome is a functional ServiceNow Transform Map configuration that automates data loading into the Institution Details table.

The completed implementation should demonstrate that:

- Bulk spreadsheet data can be staged and transformed into target ServiceNow records automatically.
- Reference fields (Student Name, Faculty Name) resolve accurately to sys_user entries.
- Choice values (ECE, EEE, CSE) map properly into target choice lists.
- Coalesce settings prevent duplicate record insertion during re-imports.
- Imported records are properly secured post-import by existing ACL rules.

12.BENEFITS

The proposed solution provides the following benefits:

- **Automation & Efficiency:** Eliminates manual record creation for large datasets.
- **Data Integrity:** Ensures references and choices conform to platform data standards.
- **Duplicate Prevention:** Coalesce logic prevents duplicate records during recurring uploads.

- **Scalability:** Easily handles high-volume bulk imports across multiple
- **Practical Implementation:** Demonstrates real-world ServiceNow data integration and ETL workflows.

13.FUTURE ENHANCEMENTS

The project can be extended with additional data import capabilities, including:

- Automated scheduled imports via FTP/MID Server.
- Advanced Transform Scripts (OnBefore, OnAfter, OnChoiceMatch).
- Complex data cleansing and string manipulation prior to target insertion.
- Automated error notification emails for failed import set rows.
- Integration with external REST APIs using Import Set API.

14.CONCLUSION

The Brainstorming and Ideation Phase establishes the foundation for the **Import Data using Transform Maps (Spreadsheets)** project. The proposed solution uses ServiceNow Import Sets, staging tables, and field mapping logic to automate bulk record creation in the Institution Details table (u_institution_details). The identified objectives, scope, and mapping structure provide the foundation for the subsequent Requirement Analysis Phase.